MILITARY SPECIFICATION

BODY ARMOR, BONNET, FOR EXPLOSIVE ORDNANCE DISPOSAL (EOD) UNITS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a body armor bonnet for protection against high velocity fragments associated with Unexploded Ordnance (UXO) and Improved Explosive Devices (IED). The insert is of aramid ballistic filler enclosed in an oxford aramid cloth shell.

1.2 Classification. The body armor bonnet shall be of one type and in the following sizes.

Sizes

Extra small
Small
Medium
Large

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8470

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SPECIFICATIONS

FEDERAL

DDD-L-20	-	Label: For Clothing, Equipage and Tentage (General Use)
PPP-B-636	-	Boxes, Shipping, Fiberboard

MILITARY

MIL-B-371	-	Braid, Textile, Tubular
MIL-T-5038	-	Tape, Textile and Webbing, Textile, Reinforcing, Nylon
MIL-C-43842	-	Cloth, Oxford, Aramid
MIL-C-44050	-	Cloth, Ballistic, Aramid
MIL-T-44100	-	Thread, Para-Aramid, Spun, Intermediate Modulus

STANDARDS

FEDERAL

FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein.

COLOR ASSOCIATION OF THE UNITED STATES, INC.

Department of Defense Standard Shades for Sewing Threads

(Application for copies should be addressed to the Color Association of the United States, Inc., 24 East 38th Street, New York, NY 10016.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case the document shall govern.

3.2 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.2 and 6.4).

3.3 Material (see 6.6).

3.3.1 Cloth, oxford, aramid. The cloth for the inner and outer shell, and the tunnel for the drawstring shall be aramid oxford, Olive Green 106, conforming to MIL-C-43842.

3.3.2 Cloth, ballistic, aramid. The cloth for ballistic filler components shall be water-repellent aramid ballistic cloth, type II, class 2 of MIL-C-44050.

3.3.3 Thread. The thread for all stitching throughout the bonnet shall be para-aramid, spun, intermediate, modulus conforming to MIL-T-44100 and shall be Olive Drab S-1, C.A. 66022. Thread size shall be as specified in table II.

3.3.4 Tape. The 1 inch wide nylon tape for joining the butted ballistic inserts shall be type IV, Olive Drab 7, conforming to MIL-T-5038.

3.3.5 Braid. The 1/8 inch textile tubular braid to be inserted into the tunnel at the bottom of the bonnet for a drawstring shall be Olive Green 107, type IV, class 2 conforming to MIL-B-371.

3.3.6 Label. Each bonnet shall have a combination identification, and size label conforming to type VI, class 4 of DDD-L-20.

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3.4 Design. The body armor bonnet shall consist of an aramid ballistic filler with aramid oxford shell fabric, of a tunnelled drawstring tightening design.

3.5 Pattern. Standard patterns to be used to cut working patterns will be furnished by the Government. The working patterns shall be identical to the standard patterns which shall not be altered in any way. The patterns provide 3/8 inch seam allowance.

3.5.1 Pattern parts. The component of the bonnet shall be cut from the materials specified below, in accordance with pattern parts indicated.

TABLE I. List of pattern parts

Material	Nomenclature	Parts
Ballistic, aramid	Crown inner	4
	Crown center	4
	Crown outer	4
	Side crown inner	8
	Side crown center	8
	Side crown outer	8
Oxford, aramid	Crown inner	1
	Crown outer	1
	Side crown inner	2
	Side crown outer	2
	Tunnel piece	1

3.6 Construction. The construction shall conform in all respects to the requirements specified in table II and herein.

3.6.1 Stitches, seams and stitching. All stitches, seams and stitching shall conform to FED-STD-751. The type of seam, stitching and stitches per inch shall be as specified in table II. Seam allowances shall be maintained with seams sewn so that no raw edges, run-offs, pleats, puckers or open seams occur. When two or more methods of seams or stitches are given for the same operation, any one may be used.

3.6.1.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread or excessively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

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3.6.1.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1 inch back of the end of the stitching. 1/

b. Thread breaks, or two or more consecutive skipped or run-off stitches noted during end item inspection shall be repaired by overstitching. The stitching shall start a minimum of 1 inch in back of the defective area, continue a minimum of 1 inch beyond the defective area onto the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. 1/

1/ When making the above repairs, the ends of the stitching are not required to be backstitched.

3.6.1.2 Type 101, 401, 502, 503, 504 and 505 stitching. Thread tension shall be maintained so that there will be no loose or excessively tight stitching. All repairs shall be in accordance with 3.6.1.1.1a and b. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.6.1.2.1 Optional type 101 holding stitching. Any additional holding stitching to facilitate manufacturing is permissible provided that during ballistic filler assembly the stay stitching is 1/4 ± 1/16 inch from edge of ballistic filler.

3.6.2 Buttonholes. The buttonholes shall be straight cut and the finished cut length shall be 1-1/4 inches long. The stitching shall be securely caught in the fabric and the ends shall be securely tacked. Additive treatments or materials used to secure the ends of the stitching or to stiffen the fabric prior to stitching the buttonholes are prohibited.

3.6.3 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch ,size and type of thread are met, and at least three or more tying, overlapping or back stitches are used to secure the ends of the stitching.

3.6.4 Thread ends. All thread ends that are visible on the finished bonnet shall be trimmed to a length of not more than 1/4 inch.

3.6.5 Lubrication of thread. Except for fabrication the ballistic filler components (see 6.5), there shall be no lubrication of the thread by any means prior to or during sewing (see 4.4.1.1).

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3.6.6 Location marks. Marks for showing location of component parts on the outer shells may be done with marking ink or by use of drill holes providing they are located such that the marks or holes are completely and permanently covered by another component part.

3.6.7 Repairs. Repairs such as mends, darns, patches or splices are not permitted on the bonnet.

3.7 Manufacturing operations requirements. The bonnet shall be manufactured in accordance with operations specified in table II. The contractor is not required to follow the exact sequence of operations.

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NO.	TABLE II. MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. All components shall be cut in strict accordance with patterns furnished which show size, cutting direction lines, placement of components, and notches for proper assembling of all parts. b. The component shall not be drilled for positioning any part except where a component part, when attached, will completely and permanently cover the drill hole. The ballistic filler may be cut in any direction. All components shall be cut in strict accordance with patterns furnished. c. Each reinforcing tape for joining ballistic filler components shall be cut in sufficient length to run the length of each seam.						
2.	Stitch ballistic fillers. a. The crown ballistic filler shall consist of 12 plies, in groups of 4 inner plies, 4 center plies, and 4 outer plies. Evenly align the plies of each group and overedge around periphery of each group with the stitch gauge 3/8 to 1/2 inch wide.	502, 503, 504, or 505	SSa-1	6-8	T-50	T-50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Stitch ballistic fillers. (cont'd)						
	b. Each side crown ballistic filler shall consist of 12 plies (left and right respectively) in groups of 4 inner plies, 4 center plies, and 4 outer plies. Evenly align the plies of each group and overedge around periphery of each group with the stitch gauge 3/8 to 1/2 inch wide.	502,503, 504, or 505	SSa-1	6-8	T-50	T-50	
3.	Assemble ballistic filler.						
	a. Join crown ballistic filler to side crown ballistic fillers (left and right respectively) within each group by butting the edges together and stitching joining tape over seam, catching both components.	301	OSa-1 or OSa-2	8-10	T-50	T-50	
	b. Position inner, center, and outer groups together and tack in at least six places each on the crown, left crown, and right crown components. Each tack shall be 1/2 to 3/4 inch long with at least three overlapping rows of stitching.	301	--	8-10	T-50	T-50	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make outer shell.						
	a. Join crown sides (right and left respectively) to crown.	301 or 401	SSa-1	8-10	T-35	T-35	
	b. With right sides up, turn seams toward crown and top stitch 1/16 to 1/8 inch from edge of seam.	301 or 401	LSd-1	8-10	T-35	T-35	
5.	Make inner shell.						
	a. Join crown sides (right and left respectively) to crown.	301 or 401	SSa-1	8-10	T-35	T-35	
	b. With right side up, turn seams towards side crown panels and top stitch 1/16 to 1/8 inch from edge of seam.	301 or 401	LSd-1	8-10	T-35	T-35	
6.	Labels.						
	The combination identification/size label shall be centered on the left crown inner shell and stitched on all four sides. Stitching shall not pass through the printing.	301	--	8-10	T-35	T-35	

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NO.	TABLE II. (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Assemble bonnet (shell fillers).						
	a. Position inner shell inside assembled ballistic filler. Smooth out the inner shell (Nomex fabric) and back tack crown/side crown seams to ballistic inserts in three to four evenly spaced places or hand tack.	301 or 101	--	8-10	T-50	T-50	
	b. Position ballistic filler inside outer shell. Stitch inner and outer shells together around periphery, not catching ballistic filler.	301	SSa-1	8-10	T-35	T-35	
8.	Attach tunnel piece.						
	a. Hem ends of tunnel piece according to marks on pattern.	301	EFb-1	8-10	T-35	T-35	
	b. With right sides together stitch tunnel piece to inside periphery of bonnet starting and ending at lower right side of label.	301	SSa-1	8-10	T-35	T-35	
	c. Turn tunnel to outside shell, insert pull cord, turn in edge, and stitch 1/16 to 1/8 inch from edge.	301	BSg-2	8-10	T-35	T-35	
	d. Evenly stitch second row of stitching 1/2 inch from bottom of tunnel containing pull cord.	301	OSa-1	8-10	T-35	T-35	
	e. Bartack tunnel strip to cover at ends.	1/4 inch bartack		21	T-35	T-35	

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TABLE II. (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Buttonholes. Stitch buttonholes on lower back bonnet according to marks on patterns. (See 3.6.2.)	Buttonhole	--	52-56 per buttonhole including tack)	T-35	T-35	
10.	Clean bonnet. Remove all spots, stains, and ends of thread.						

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3.8 Finished measurements. The finished measurements shall conform to measurements for each respective size, as shown in table III.

TABLE III. Finished measurements (inches)

Sizes	X-Small	Small	Medium	Large	Tolerance
Front to back 1/	18-1/2	19-1/2	20-1/2	21	± 1/2
Side to side 2/	19-1/2	20-1/2	21-1/2	22	± 1/2

1/ Measure center front to center back from stitching line to stitching line.

2/ Measure side to side at longest points from stitching line to stitching line.

3.9 Workmanship. The finished body armor bonnet shall conform to the quality of product established by this document. The appearance of defects shall not exceed the specified quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specifed, in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless otherwise disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. When certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the thread lubrication restriction requirement specified in 3.6.5.

4.4.2 In-process inspection.

4.4.2.1 In-process examination of cut parts, fillers and patterns. All cut parts for the shell and filler assemblies shall be examined during the cutting process to determine that parts are cut properly with respect to size and material directional requirements, and that parts containing defects such as a hole, cut, smash, float, loose slub, hard crease or wrinkle are removed from production. All ballistic fillers shall be examined during assembly of the individual groups to assure that they contain the correct amount of plies. In addition to the above, inspection shall be made of the working patterns to asure that they conform to the Government patterns in all respects. Whenever nonconformance is noted, correction shall be made to the item or items affected. Parts which cannot be corrected shall be removed from production.

4.4.2.2 In-process size examination for filler groups. The ballistic filler groups after edge stitching shall be examined for the defects listed below. The lot size shall be expressed in terms of left, right, or center filler groups, as applicable. The sample unit shall be one left, right, or center filler group, as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 1.0 for major defects and 4.0 for total (major and minor combined) defects.

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Examine	Defect	Classification Major	Minor
Size of individual ballistic filler	a. Smaller than cutting pattern		
	(1) 3/16 inch to 3/8 inch at any point around the periphery 1/ 2/		X
	(2) More than 3/8 inch at any point around the periphery 1/ 2/	X	
	(3) 1/8 inch up to 3/16 inch around entire periphery 2/		X
	(4) More than 3/16 inch around entire periphery 2/	X	
	b. Larger than cutting pattern by 3/8 inch or more at any point 1/ 2/		X

1/ To be scored when condition exists for a length of more than 2 inches or if condition exists in several areas with an accumulated distance of 4 inches or more.

2/ The individual component, as applicable, shall be examined with the applicable cutting pattern centered on the filler component.

4.4.3 End item visual examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of bonnets. The sample unit shall be one completed bonnet. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 15.0 for total (major and minor combined) defects.

Examine	Defect	Classification Major	Minor
Cloth	a. Any hole, (except drill hole) cut, or tear.	X	
	b. Any exposed drill hole.		X
	c. Any abrasion marks, broken or missing yarns, or multiple floats.	X	
	d. Any mend, darn, or patch.	X	

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Examine	Defect	Classification Major	Classification Minor
Cloth (cont'd)	e. Parts not cut correctly with respect to material lines.	X	
	f. Needle chews.	X	
Tape	a. Any hole, cut, or frayed edge.	X	
	b. Needle chews:		
	(1) Up to 1/8 inch in length.		X
	(2) 1/8 inch or more in length.	X	
	c. Not continuous, i.e., spliced.	X	
Seams and stitchings:			
Open seams	a. Up to and including 1/2 inch.		X
	b. More than 1/2 inch.	X	
NOTE:	A seam shall be classified as an open seam when one or more stitches joining a seam are broken, or when two or more consecutive skipped or runoff stitches occur.		
Raw edges (except where required or otherwise classified herein)	More than 1/2 inch when securely caught in stitching.		X
NOTE:	Raw edges not securely caught in stitching shall be classified as open seams.		
Seam and stitch type	a. Wrong seam or stitch type.	X	
	b. Seam pleated or badly puckered, clearly noticeable.		X
Stitch tension	a. Tension loose, resulting in loose bobbin or top thread.		X
	b. Tension tight, resulting in puckering of material.		X

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Examine	Defect	Classification Major	Classification Minor
Seams and stitchings: (cont'd)			
Stitches per inch	a. One to two stitches less than minimum specified. 1/		X
	b. Three or more stitches less than minimum specified. 1/	X	
	c. One or more stitches in excess of maximum specified. 1/		X

1/ Defects to be scored only when condition exists for a length of 3 inches or more or in several areas with an accumulated distance of 5 inches or more. Applicable to individual seams.

NOTE: Variation in the number of stitches per inch caused by operator speeding up the machine and pulling the material in order to sew over heavy places, or in turning corners shall be classified as follows:

(a) Within the minor defect classification - No defect

(b) Within the major defect classification - Minor defect

Thread break, or skipped stitches (on type 301 stitching)	Overstitched less than 1 inch in each direction beyond defective stitching area.		X

NOTE: On all types of stitching, thread breaks or two or more consecutive skipped stitches not overstitched shall be classified as open seam.

Stitching ends (on type 301 stitching)	Ends of stitching not secured as specified (except when caught in other stitching or turned under in a hem).		X

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Examine	Defect	Classification Major	Minor
Runoffs	a. On joining or closing seams - use open seam classification.		
	b. Edge or raise stitching, not resulting in any open seam, more than 1/2 inch in length (when resulting in an open seam use open seam classification.)		X
Rows of stitching	Any row missing (unless otherwise classified herein).	X	
Assembly- general	a. Any component caught in any unrelated operations of stitching, affecting appearance or function (not otherwise classified herein.)	X	
	b. Any component part omitted (not otherwise classified herein) or not attached as specified.	X	
	c. Any row of stitching securing ballistic filler to inner shell omitted.	X	
Labels, (size, and identification)	a. Missing, incorrect, or illegible.	X	
	b. Misplaced.		X
Cleanness	a. Any spot or stain clearly noticeable.		X
	b. Thread ends not trimmed as specified.		X
Color of components	Color of any component not as specified.		X

4.4.4 End item dimensional examination. The end item shall be examined for conformance to the dimensions specified in table III. Any dimension deviating from the specified requirement shall be classified as a defect. The lot size shall be expressed in units of bonnets. The sample unit shall be one bonnet. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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4.4.5 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged, except that it need not be closed. Examination for the closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of shipping containers in the inspection lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: not properly secured, incomplete closure of box flap, improper taping or tying, or inadequate stapling. Bulged or distorted box and container.
Content	Number of items per shipping container is more or less than required.

4.4.6 Palletization inspection. An inspection shall be made to determine that the palletization complies with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one palletized unit load fully packaged. The lot size shall be the number of palletized unit loads in the inspection lot. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirements.
Palletization	Pallet pattern not as specified. Interlocking of loads not as specified. Load not bonded with required straps as specified.
Weight	Exceeds maximum load limits.
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Five bonnets, of one size only, shall be evenly stacked with the inside facing up. The stack of five bonnets shall be inserted into a close-fitting clear polyethylene film bag having a thickness of 0.0015 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous, and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. A 1/4-inch diameter hole shall be made at one corner of the bag to allow excess air to escape.

5.1.2 Commercial. The bonnets shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Thirty bonnets of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC-L, grade V2s of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class weather-resistant, variety DW, grade V15c of PPP-B-636. Each fiberboard shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance 4.4.5.

5.2.2 Level B packing. Thirty bonnets of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to RSC-L, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. The inside of each shipping container shall be fitted with a box liner conforming to type CF, class domestic, variety DW, grade 275 of PPP-B-636. Each fiberboard shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.5.

5.2.3 Commercial packing. Bonnets preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

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5.3 Palletization. When specified (see 6.2), bonnets, packed as specified in 5.2, shall be palletized in accordance with load type I of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L. Pallet patterns shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. The body armor is intended for use for protection against high velocity fragments associated with Unexploded Ordance (UXO) and Improvised Explosive Devices (IED).

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document
b. Size required (see 1.2).
c. When a first article sample is required (see 3.2, 4.3 and 6.4).
d. Selection of the applicable level of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 Standard samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one completed body armor assembly as designated by the procuring activity. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Cooling of needle. In order to keep the needle cool when overedging the ballistic fillers, it is recommended that either the machines be equipped with a compressed air blower directed at the needle or by affixing a felt pad saturated with silicone oil in a position so that the thread will rub against it during sewing. The use of chromium plated needles have been found to be advantageous in that they do not generate as much heat or deflect as much during use as regular needles. Recommended maximum speed for the overedge machine is 1000 RPM.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the document (see 3.3).

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Air Force - 99

Preparing activity:

Army - GL

Project No. 8470-0096

Review activities:

Army - MD
DLA - CT
Air Force - 82

User activity:

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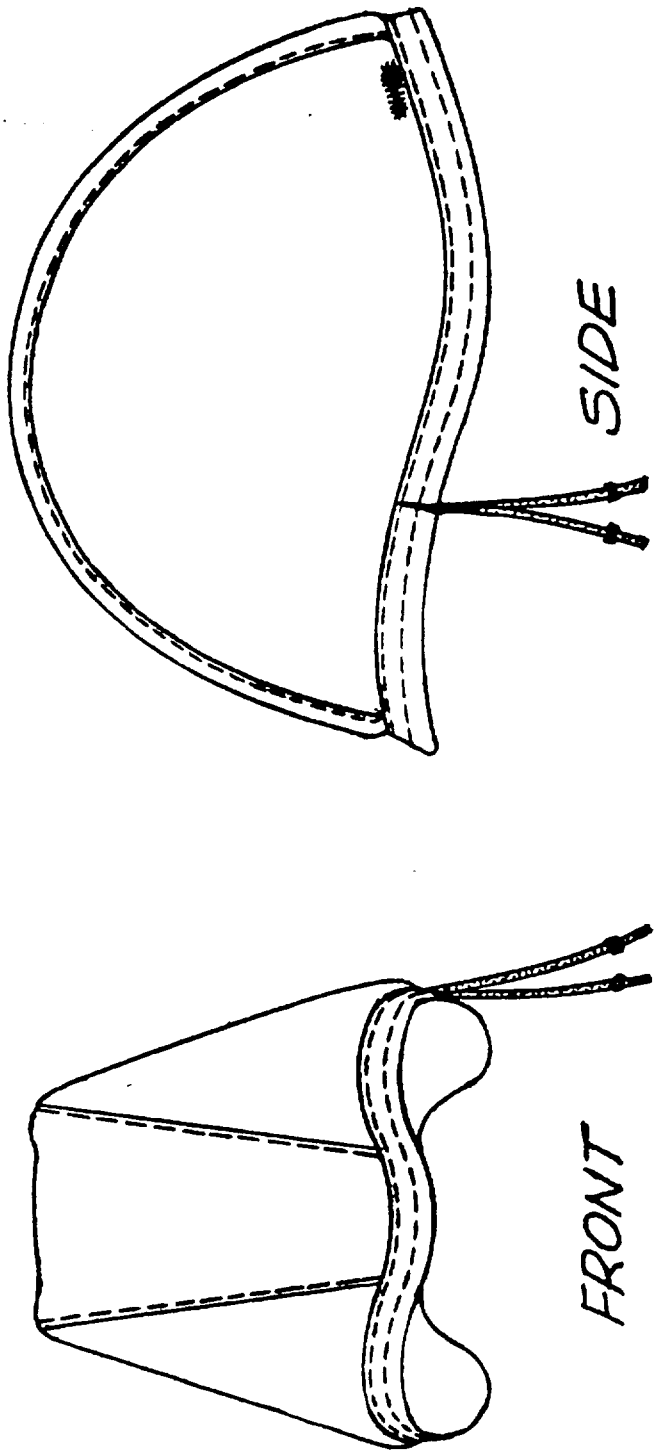


FIG. 1 BONNET, BODY ARMOR FOR (EOD)

2-1-1618

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-44127	Body Armor, Bonnet, for Explosive Ordnance Disposal (EOD) Units

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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